

Unnamed_Unit

Unit 2 Bringing Ideas to Life Face Expression Device with Joystick

PDF Documents

PDF Document 1: 1731843162948-Chapter 4_ unit2 Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.



UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

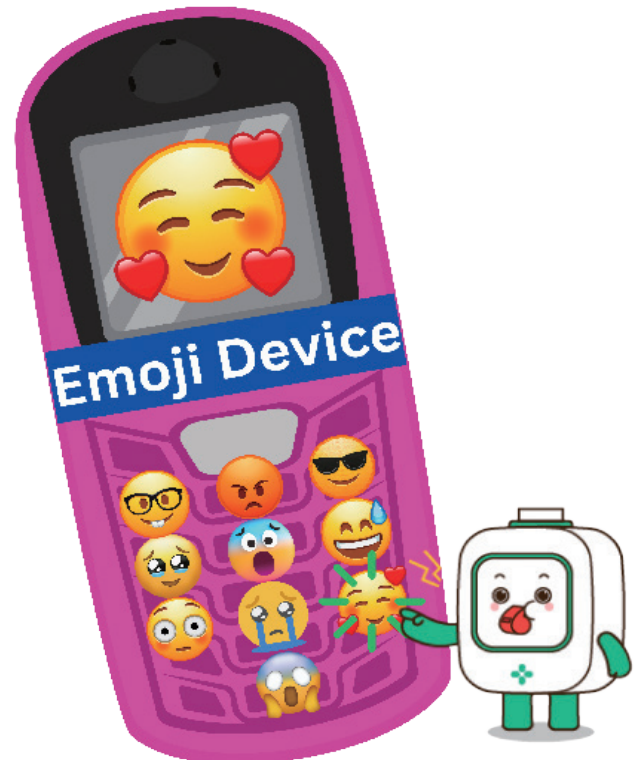
FACE EXPRESSION DEVICE FOR ALEX

In class, everyone's buzzing about the school science fair, and Cathy has a brilliant idea. She's noticed that their friend Alex is a bit shy and sometimes finds it hard to say how she's feeling, especially in big groups. Cathy thinks, "What if we could make something to help Alex express herself without having to say a word?"

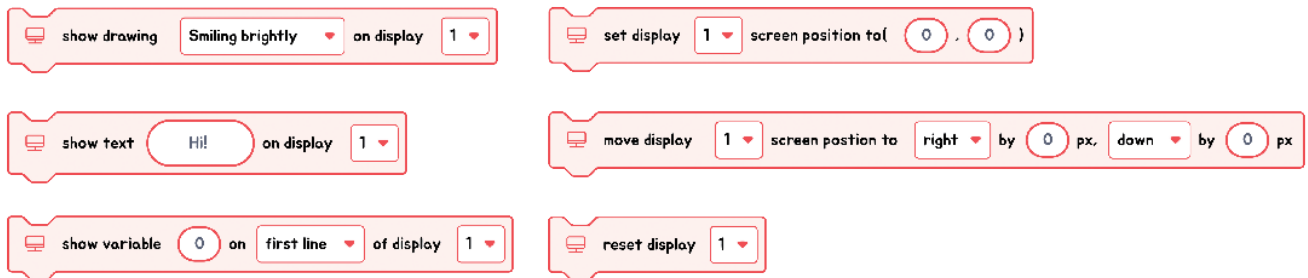


So, Cathy and her classmates decide to build a "Face Expression Device" using their MODI kit. They plan to design a gadget that can show different emojis based on how Alex is feeling. If Alex is happy, she can show a smiling emoji. If she's feeling a bit down or needs a break, he could choose a sad or a "tired" emoji.

Their goal is to give Alex a fun and easy way to share her mood with the class, making it easier for her to join in the conversation without feeling overwhelmed. Plus, they think it'll be a cool project that combines art, technology, and a bit of heart – exactly what the science fair is all about!



ABOUT DISPLAY MODULE

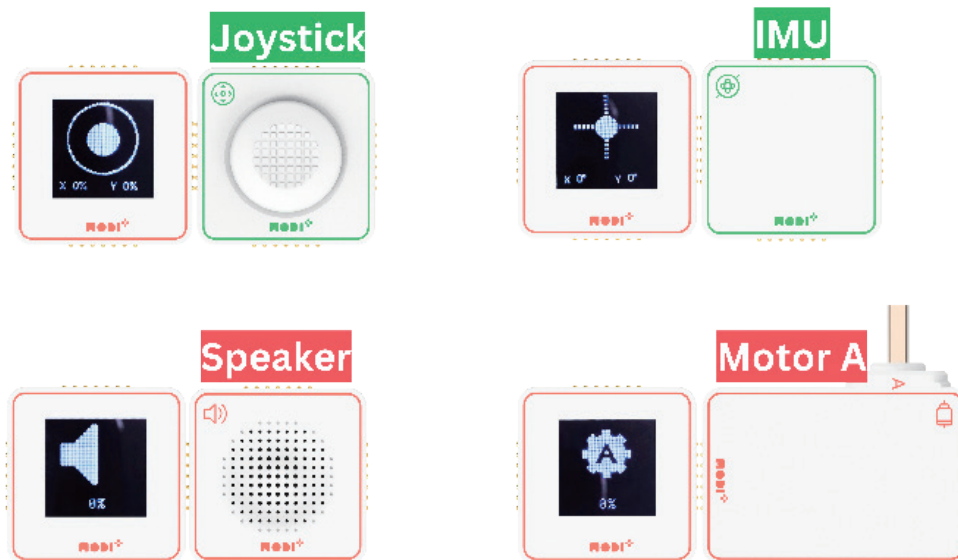


The Display module is like a tiny screen for your MODI projects. It uses 6 different blocks in the Code Editor to show pictures or information from other MODI modules nearby. So, when other modules talk to the Display, it can show you what they're saying or doing with images or data right on its screen!

AUTOMATIC DATA DISPLAY WITH MODI MODULES

The Display module is like the heart of a conversation among MODI modules. Imagine it as a screen that lights up with information whenever other modules, like the Button or the Light Sensor, have something to "say." When any input or output module is connected to it, the Display module automatically picks up their signals and shows what they're doing on its screen.

INTERACTION BETWEEN DISPLAY AND OTHER MODULES

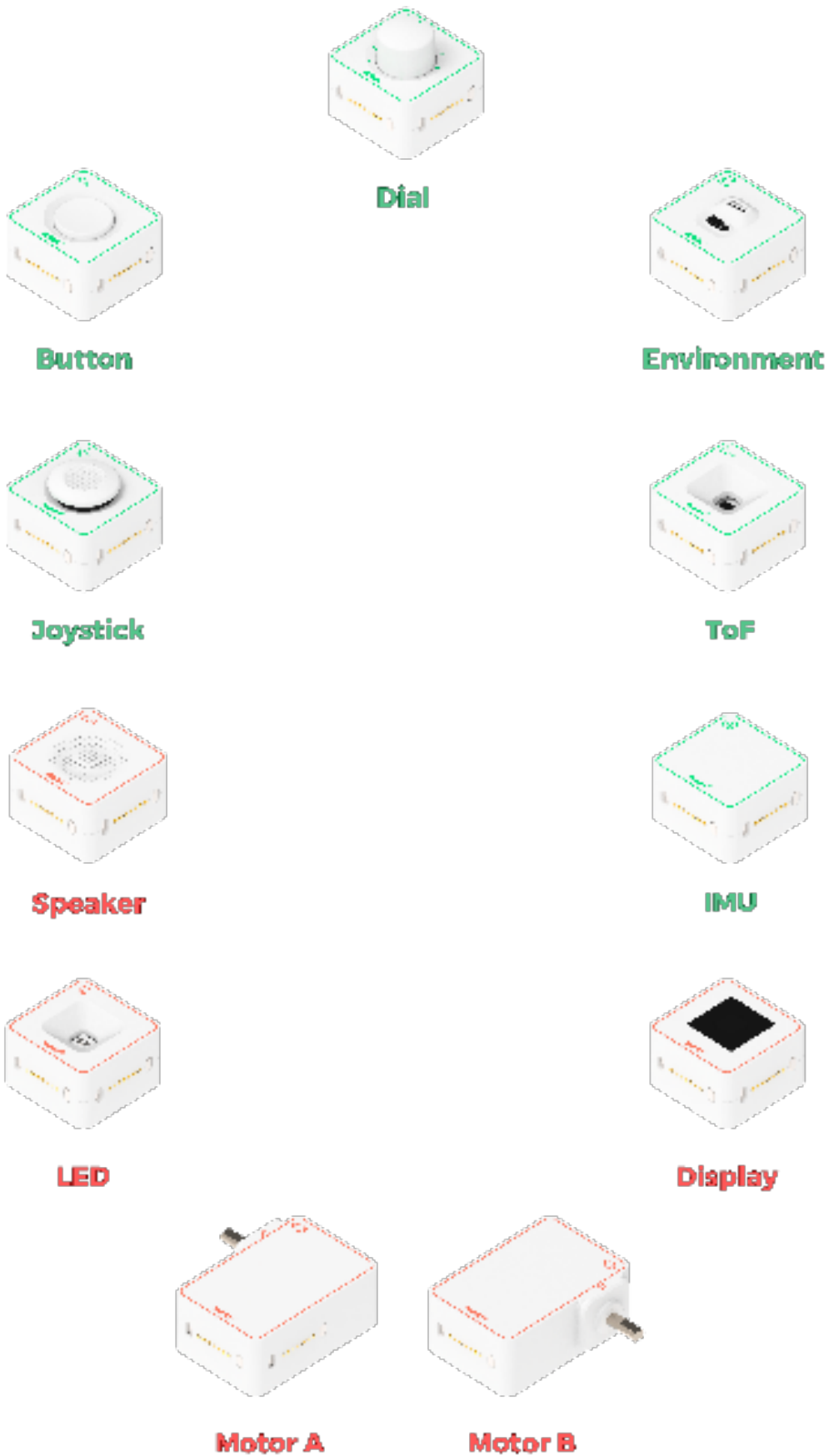


So, if you press a Button module or cover a Light Sensor, the Display will show these actions without needing any special programming. It's like having a translator that instantly shows you what's happening in the MODI module language!

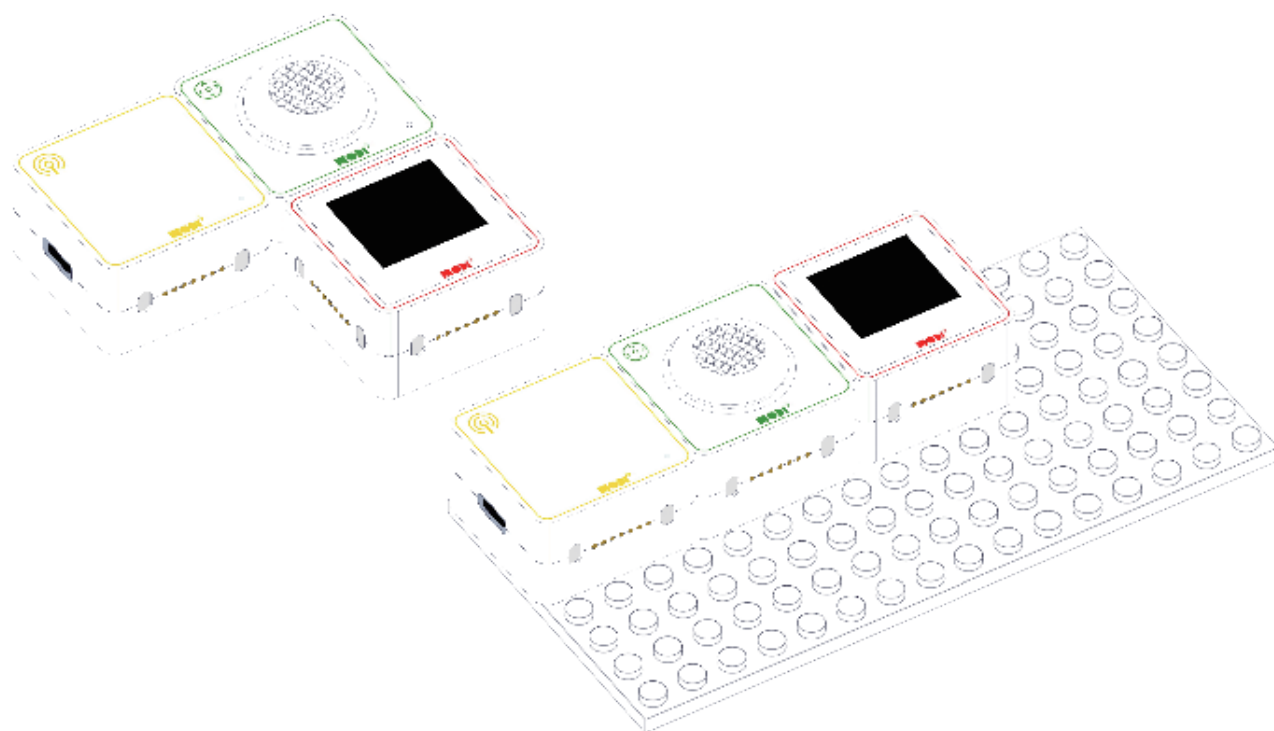


Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



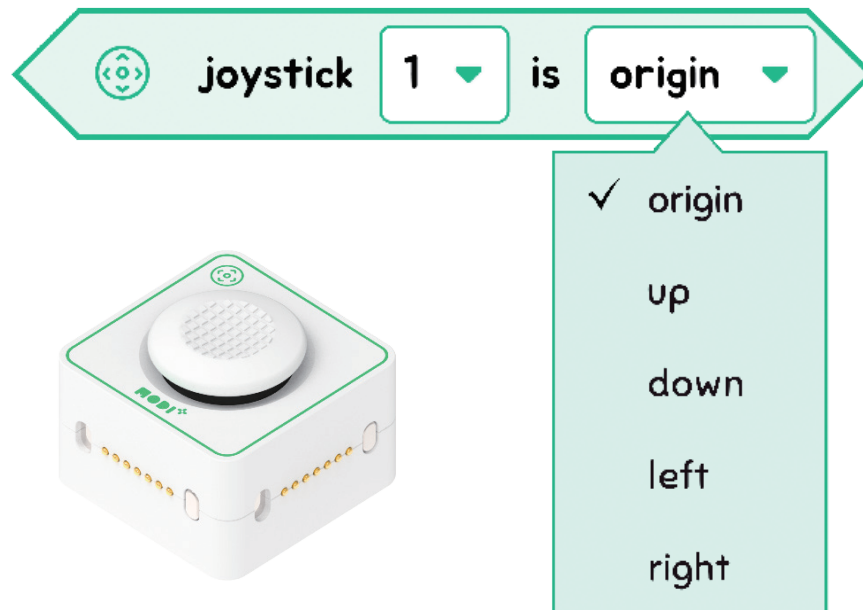
HOW DOES THE CREATION LOOK LIKE?



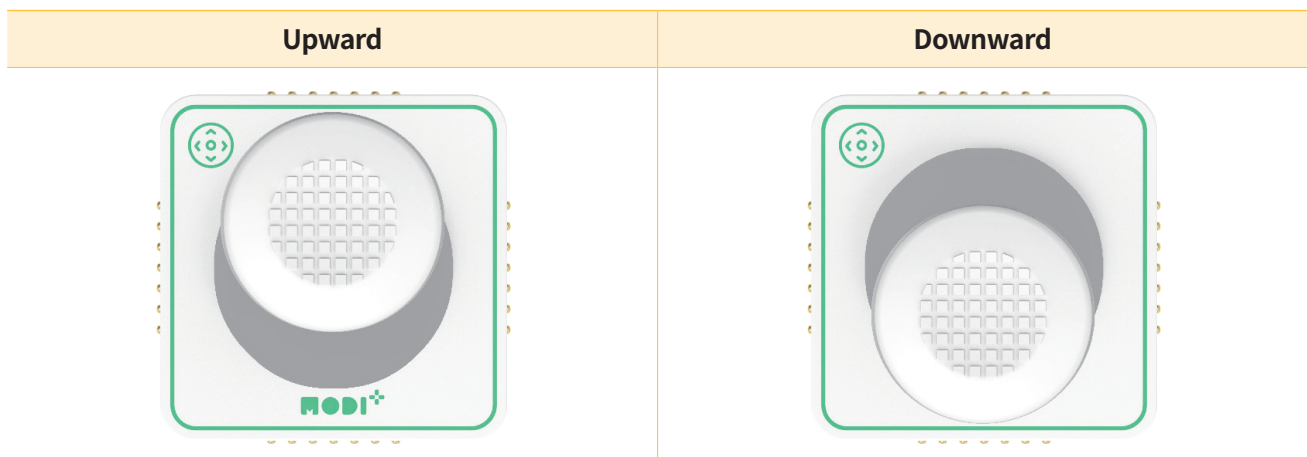


Module Functionality

Let's look into the details of the Joystick module.



Including the neutral position 'origin,' the joystick has 5 different positions.



Pushing it upward and downward affects movement on the **y-axis**
 - Range : -100 % (fully downward) ~ +100 % (fully upward)



Moving the joystick left and right controls movement along the **x-axis**